UFACTORY xArm 6 vs Universal Robots UR5e

Comparison for robotic integration in a pharmaceutical facility | Created 2026-05-28

Purpose

This document compares the UFACTORY xArm 6 and Universal Robots UR5e from a practical integration perspective: robot capability, controller/interface, programming workflow, safety evidence, and pharmaceutical-facility concerns. It is not a formal safety validation or GMP qualification document.

Executive summary

Decision point	Practical answer
Best for regulated production cell	UR5e is the stronger default because Universal Robots provides clearer industrial safety documentation, including PL d / Category 3 evidence for e-Series safety functions and safety I/O [3-5].
Best for low-cost prototype/research	xArm 6 is lower-cost and convenient for PC/browser, Python, C++, ROS, and ROS 2 workflows, but the safety evidence found is less explicit than UR's certified safety-function documentation [1,2].
Pharma-specific issue	Do not decide only by payload and reach. The full cell includes robot, end-effector, controller, software, sensors, power, I/O, guarding, cleaning, maintenance, and records [6,7].
Critical risk	'Collaborative robot' does not automatically mean safe for open human operation. A task-specific risk assessment is still required.

Main comparison

Attribute	UFACTORY xArm 6	Universal Robots UR5e
Robot type	6-axis collaborative robotic arm [1]	6-axis collaborative robotic arm [3]
Payload	5 kg [1]	5 kg / 11 lb [3]
Reach	700 mm [1]	850 mm / 33.5 in [3]
Repeatability	+/-0.1 mm [1]	+/-0.03 mm per ISO 9283 [3]
Robot-arm weight	12.2 kg [1]	20.6-20.7 kg, depending on source revision/datasheet [3]
Maximum TCP speed	1 m/s [1]	Approx. 4 m/s in current UR technical specification; older datasheets emphasize application settings and safety-limited operation [3]
Controller	External xArm control box required; robot arm hardware composition includes robotic arm and control box [1].	External UR control box required; UR system uses control box plus teach pendant [3].
Main interface	UFACTORY Studio through PC/browser/control-box IP; supports Live Control, Blockly, Python IDE, G-code/M-code [2].	PolyScope graphical interface on 12-inch touchscreen teach pendant [3].
External programming	UFactory Studio, Python, C++, ROS, private TCP, Modbus TCP options are documented in UFACTORY materials [1,2].	PolyScope, URScript, RTDE/client-library workflows, ROS, and ROS 2 drivers are supported in UR ecosystem/documentation [3].
Cleanroom statement	xArm common specs list ISO Class Cleanroom 5 [1].	UR5e datasheet lists ISO 14644-1 Class 5 cleanroom classification at specified conditions [3].
Safety boundary / workspace limiting	Safety Boundary limits Cartesian working range; if TCP exceeds boundary, robot stops [1].	Configurable safety functions include position limits/planes and safety I/O in the UR safety system [4,5].
Reduced mode	Reduced Mode limits maximum linear speed, joint speed, and joint range [1].	Reduced mode and other safety functions are part of UR e-Series safety configuration [4,5].

Attribute	UFACTORY xArm 6	Universal Robots UR5e
Emergency stop	Control-box E-stop stops activities, clears cached commands, and removes robot-arm power within 300 ms; manual states E-stop should not be used as the risk-reduction measure [1].	UR e-Series E-stop is documented as a safety function; UR e-Series safety functions and safety I/O are PL d / Category 3 according to ISO 13849-1 [4,5].
Collision detection	Collision Detection, Collision Rebound, and Self-collision Detection are documented in xArm advanced settings [1].	UR safety/control ecosystem includes configurable safety functions and protective stop behavior; certification evidence is stronger in official UR safety documents [4,5].
Safety certification evidence found	The reviewed UFACTORY manuals document features and warnings, but I did not find a clear official statement equivalent to UR's PL d / Category 3 certification for safety functions and safety I/O.	Official UR source states e-Series safety functions and safety I/O are PL d / Category 3 according to ISO 13849-1, with TUV NORD certificate #44 207 14097610 [4].
Integration maturity	Good for research, lab automation, and lower-cost applications; pharma deployment needs strict external safety and validation work.	More mature for industrial integration, safety documentation, teach-pendant workflow, and validated automation environments.

Safety and compliance interpretation

For a pharmaceutical facility, the important question is not only whether the robot is collaborative. The complete robot cell must be assessed: manipulator, end-effector, controller, software, sensors, power source, I/O, guarding, cleaning access, maintenance, and records. OSHA describes an industrial robot system as including the robot, end-effector, control system, power sources, sensors, and communication/I/O devices [6].

Topic	xArm 6 implication	UR5e implication	Pharma action
Risk assessment	Manual gives safety precautions and safety features, but cell-level validation remains integrator responsibility [1].	Better documented safety-function evidence; still requires task-specific risk assessment [4,5].	Perform formal risk assessment for robot, gripper, fixtures, product handling, operator access, cleaning, and maintenance.
Emergency stop chain	Control-box E-stop removes robot-arm power within 300 ms [1].	E-stop is part of documented safety-function architecture [4,5].	Design and test an E-stop circuit for the entire cell, not only the robot arm.
End-effector hazard	Gripper/vacuum/tool is external to the arm and can dominate risk [1,6].	Same issue; UR certification does not automatically certify the custom tool.	Validate pinch, dropped-part, sharp-edge, contamination, and cleaning risks for the end-effector.
Cleanability	Robot may be listed as ISO Class 5, but cables, grippers, covers, lubricants, and cleaning chemicals must be checked [1].	UR5e datasheet includes cleanroom classification, but final cell cleanability still depends on full setup [3].	Confirm compatibility with disinfectants, cleanroom class, cable routing, particle generation, and wipe-down procedures.
Software records	Studio/Blockly/Python programs may need controlled backup and versioning [2].	PolyScope/URScript/RTDE programs also need controlled backup/versioning.	If electronic records or GMP-critical data are created/maintained electronically, evaluate 21 CFR Part 11 applicability [8].
Qualification	Likely requires more custom documentation and verification.	Usually easier to justify due to mature safety documentation.	Define URS, risk assessment, IQ/OQ/PQ, change control, backup/restore, access control, and maintenance plan.

Recommendation

For a fixed pharmaceutical automation cell, choose UR5e if budget allows and safety/compliance documentation is a major concern. Choose xArm 6 mainly for prototype, research, low-risk non-aseptic handling, or cost-sensitive automation where the integrator can add and validate external safety hardware, guarding/interlocks, software controls, and cleaning/maintenance procedures.

> Do not rely on built-in robot safety features alone. The validated system must include the robot, controller, end-effector, safety PLC or safety relay if required, guarding or presence-sensing devices if required, network/security design, software backup/restore, operator procedure, cleaning SOP, maintenance SOP, and change-control process.

Minimum pharma integration checklist

Area	Minimum checks before implementation
Application definition	Define product contact vs non-contact, aseptic vs non-aseptic, batch-critical vs support-only, payload, reach, speed, cycle time, and failure modes.

Area	Minimum checks before implementation
Safety	Perform task-based risk assessment; validate E-stop, protective stop, interlocks, reduced mode, guarding, and recovery procedure.
End-effector	Assess pinch/crush/shear hazards, dropped-load risks, material compatibility, cleanability, and product contamination risk.
Cleaning/GMP	Confirm equipment design supports operation, cleaning, and maintenance as required by 21 CFR 211.63 [7].
Software/control	Control program versioning, backup/restore, access control, audit trail expectations, and validation scope. Consider 21 CFR Part 11 if electronic records/signatures are GMP-relevant [8].
Documentation	Generate URS, design specification, wiring diagrams, safety validation, IQ/OQ/PQ protocol, maintenance plan, calibration/inspection plan, and change control.

References

[1] UFACTORY, xArm User Manual V2.0.0, hardware section and appendix technical specifications.

[2] UFACTORY, UFACTORY Studio User Manual V2.6.0, browser-based Studio interface, Live Control, Blockly, Python IDE, and G-code/M-code sections.

[3] Universal Robots, UR5e e-Series Technical Specification / Datasheet, updated 2024.

[4] Universal Robots, UR e-Series Functional Safety: Safety Functions and Safety I/O are PL d, Category 3 according to ISO 13849-1, TUV NORD certificate #44 207 14097610.

[5] Universal Robots, Safety Functions Table 1, PL d Category 3 safety functions and safety I/O, PFH_D statement.

[6] OSHA Technical Manual, Section IV, Chapter 4: Industrial robot systems include robot, end-effector, control system, power sources, sensors, and I/O devices.

[7] 21 CFR 211.63, Equipment design, size, and location: equipment must facilitate intended operation, cleaning, and maintenance.

[8] FDA Guidance for Industry: Part 11, Electronic Records; Electronic Signatures - Scope and Application.

Source links

UFACTORY xArm User Manual V2.0.0: https://www.ufactory.cc/wp-content/uploads/2023/05/xArm-User-Manual-V2.0.0.pdf

UFACTORY Studio User Manual V2.6.0: https://www.ufactory.cc/wp-content/uploads/2025/05/UFACTORY_Studio_User_Manual_V2.6.0.pdf

UR5e Technical Specifications: https://www.universal-robots.com/media/1807465/ur5e_e-series_datasheets_web.pdf

UR e-Series Functional Safety: https://www.universal-robots.com/media/1804320/e-series-functional-safety.pdf

UR Safety Functions Table:
https://www.universal-robots.com/manuals/EN/HTML/SW10_6/Content/prod-usr-man/hardware/arm_e-Series/UR5e/H_g5_sections/safetyFunctionsAndinterfaces/safety_functions_table1_g5_en.htm

OSHA Technical Manual, Industrial Robot Systems: https://www.osha.gov/otm/section-4-safety-hazards/chapter-4

21 CFR 211.63: https://www.ecfr.gov/current/title-21/chapter-I/subchapter-C/part-211/subpart-D/section-211.63

FDA Part 11 Guidance: https://www.fda.gov/regulatory-information/search-fda-guidance-documents/part-11-electronic-records-electronic-signatures-scope-and-application